1. DESCRIPTIVE STATISTICS:

BEFORE

HEIGHT		*WEIGHT*	
Mean	162.862	Mean	59.3929
Standard Error	0.9672701	Standard Error	1.97612
Median	164.295	Median	55
Mode	160	Mode	45
Standard Deviation	8.0927623	Standard Deviation	16.5334
Sample Variance	65.492802	Sample Variance	273.354
Kurtosis	0.7975491	Kurtosis	1.49987
Skewness	-0.222195	Skewness	1.13158
Range	43.18	Range	80
Minimum	137.16	Minimum	35
Maximum	180.34	Maximum	115
Sum	11400.34	Sum	4157.5
Count	70	Count	70

AFTER

HEIGHT		*WEIGHT*	
Mean	166.12014	Mean	60.6857
Standard Error	1.0393455	Standard Error	1.51426
Median	165	Median	59.5
Mode	165	Mode	54
Standard Deviation	8.695788	Standard Deviation	12.6692
Sample Variance	75.616729	Sample Variance	160.508
Kurtosis	0.4915407	Kurtosis	1.82806
Skewness	-0.433064	Skewness	0.92049
Range	44.84	Range	70
Minimum	137.16	Minimum	38
Maximum	182	Maximum	108
Sum	11628.41	Sum	4248
Count	70	Count	70

2. CORRELATION:

S.NO.	BEFORE HEIGHT (IN CM)	BEFORE WEIGHT (IN KG)	AFTER HEIGHT (IN CM)	AFTER WEIGHT (IN KG)
1	158	40	161	45
2	155	73	157	71
3	160	75	165	70
4	162	84	167	76
5	173.73	86.5	173.73	72
6	166	62	168	67
7	160	55	173	65
8	168	47	168	51
9	164	56	164	52
10	160	75	160	75
11	167.64	52	173.73	54
12	158.49	38	164.59	47
13	148	67	152	78
14	164.59	68	167.64	63
15	154	84	154	60
16	165	55	171	58
17	155.75	58	155.48	62
18	164.59	60	164.59	59
19	164.59	45	173.73	55
20	168	50	180	68
21	157.48	54	157.48	53
22	160	60	163	57
23	167.64	55	176.78	65

CORRELATION BEFORE
0.366312

CORRELATION AFTER
0.451857

Sheet5 | Sheet1 | DESCRIPTIVE | CORRELATION | T TEST PAIRED FEMALE | T TEST PAIRED

3. Z-TEST: TWO SAMPLE FOR MEANS

- FEMALE (HEIGHT)

z-Test: Two Sample for Means

	HEIGHT BEFORE	*HEIGHT AFTER*
Mean	158.96	161.25
Known Variance	43.00	52.54
Observations	34.00	34.00
Hypothesized Mean	0.00	
z	-1.37	
P(Z<=z) one-tail	0.09	
z Critical one-tail	1.64	
P(Z<=z) two-tail	0.17	
z Critical two-tail	1.96	

B	C	D
BEFOERE HEIGHT (IN CM)	**AFTER HEIGHT (IN CM)**	**GENDER**
158	161	FEMALE
155	157	FEMALE
160	165	FEMALE
168	168	FEMALE
164	164	FEMALE
160	160	FEMALE
167.64	173.73	FEMALE
158.49	164.59	FEMALE
154	154	FEMALE
155.75	155.48	FEMALE
164.59	164.59	FEMALE
164.59	173.73	FEMALE
157.48	157.48	FEMALE
160	163	FEMALE
164.59	164.59	FEMALE
161.54	162	FEMALE
167.64	168	FEMALE
152	152	FEMALE
164.59	170.68	FEMALE
160	165	FEMALE
165	165	FEMALE
152	154	FEMALE
150	152	FEMALE
162	162	FEMALE
155.44	161.54	FEMALE

- **FEMALE WEIGHT**

z-Test: Two Sample for Means

	BEFORE WEIGHT	*AFTER WEIGHT*
Mean	52.79	53.21
Known Variance	150.16	73.01
Observations	34.00	34.00
Hypothesized Mean Difference	0.00	
z	-0.16	
P(Z<=z) one-tail	0.44	
z Critical one-tail	1.64	
P(Z<=z) two-tail	0.87	
z Critical two-tail	1.96	

BEFORE WEIGHT (IN KG)	AFTER WEIGHT (IN KG)	GENDER
40	45	FEMALE
73	71	FEMALE
75	70	FEMALE
47	51	FEMALE
56	52	FEMALE
75	75	FEMALE
52	54	FEMALE
38	47	FEMALE
84	60	FEMALE
58	62	FEMALE
60	59	FEMALE
45	55	FEMALE
54	53	FEMALE
60	57	FEMALE
54	53	FEMALE
54	54	FEMALE
40	45	FEMALE
52	48	FEMALE
45	50	FEMALE
55	60	FEMALE
61	58	FEMALE
40	47	FEMALE

MALE HEIGHT

z-Test: Two Sample for Means

	BEFORE HEIGHT	AFTER HEIGHT
Mean	166.55	170.72
Known Variance	59.75	54.70
Observations	36.00	36.00
Hypothesized Mean Difference	0.00	
z	-2.34	
P(Z<=z) one-tail	0.01	
z Critical one-tail	1.64	
P(Z<=z) two-tail	0.02	
z Critical two-tail	1.96	

BEFORE HEIGHT (IN CM)	AFTER HEIGHT (IN CM)	GENDER
162	167	MALE
173.73	173.73	MALE
166	168	MALE
160	173	MALE
148	152	MALE
164.59	167.64	MALE
165	171	MALE
168	180	MALE
167.64	176.78	MALE
167.64	170.68	MALE
175	175	MALE
156	160	MALE
167	170	MALE
155	155	MALE
155.44	179.83	MALE
175	175	MALE
165	165	MALE
158.49	159.34	MALE
160	166	MALE
168	168	MALE
165	165	MALE

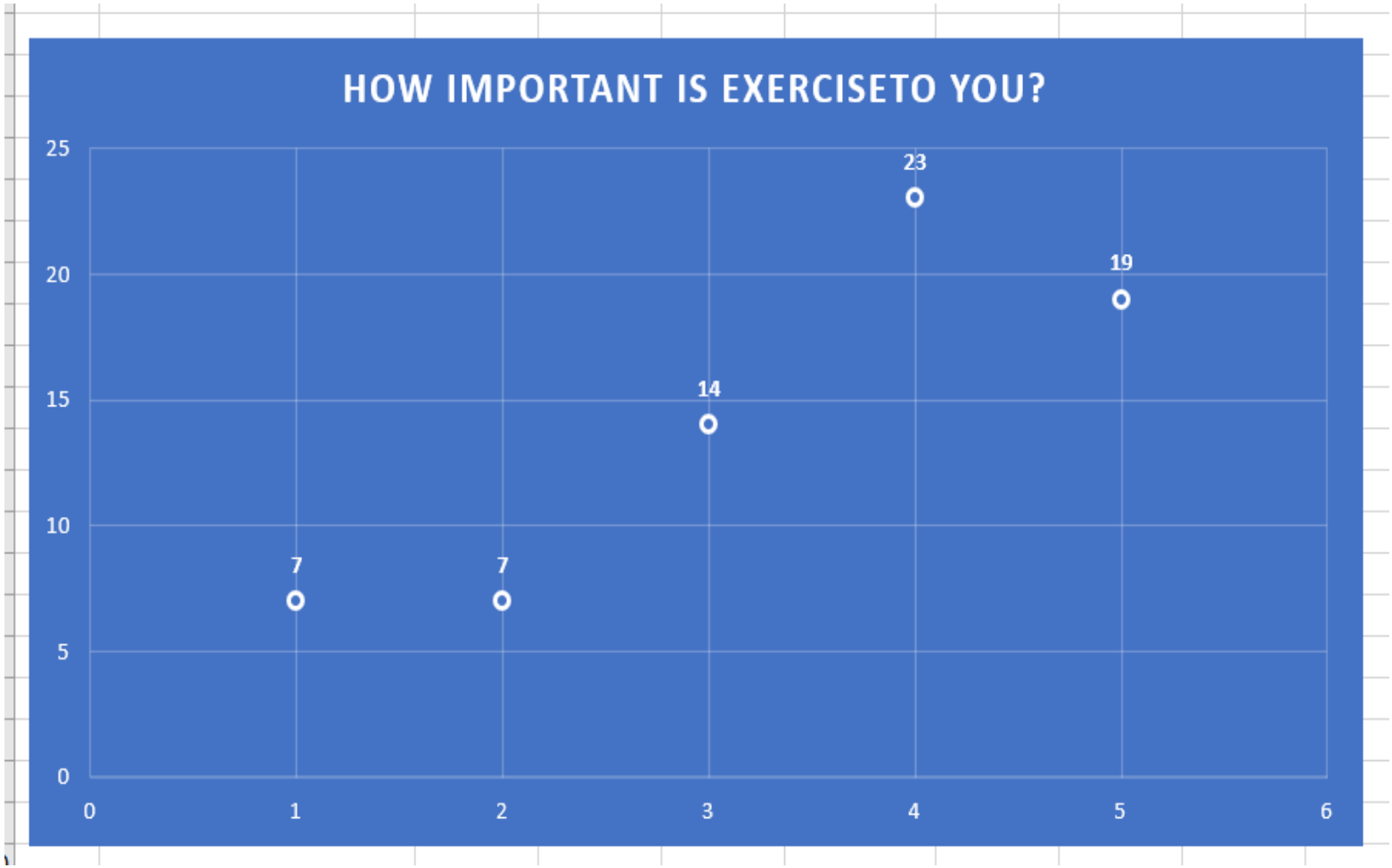
MALE WEIGHT

z-Test: Two Sample for Means

	WEIGHT BEFORE	*WEIGHT AFTER*
Mean	65.10	67.51
Known Variance	315.06	141.90
Observations	35.00	35.00
Hypothesized Mean Difference	0.00	
z	-0.67	
P(Z<=z) one-tail	0.25	
z Critical one-tail	1.64	
P(Z<=z) two-tail	0.50	
z Critical two-tail	1.96	

BEFORE WEIGHT (IN KG)	AFTER WEIGHT (IN KG)	GENDER
84	76	MALE
86.5	72	MALE
62	67	MALE
55	65	MALE
67	78	MALE
68	63	MALE
55	58	MALE
50	68	MALE
55	65	MALE
49	54	MALE
50	62	MALE
50	55	MALE
79	71	MALE
50	50	MALE
70	65	MALE
73	68	MALE
72	85	MALE
51	51	MALE
115	93	MALE
52	68	MALE

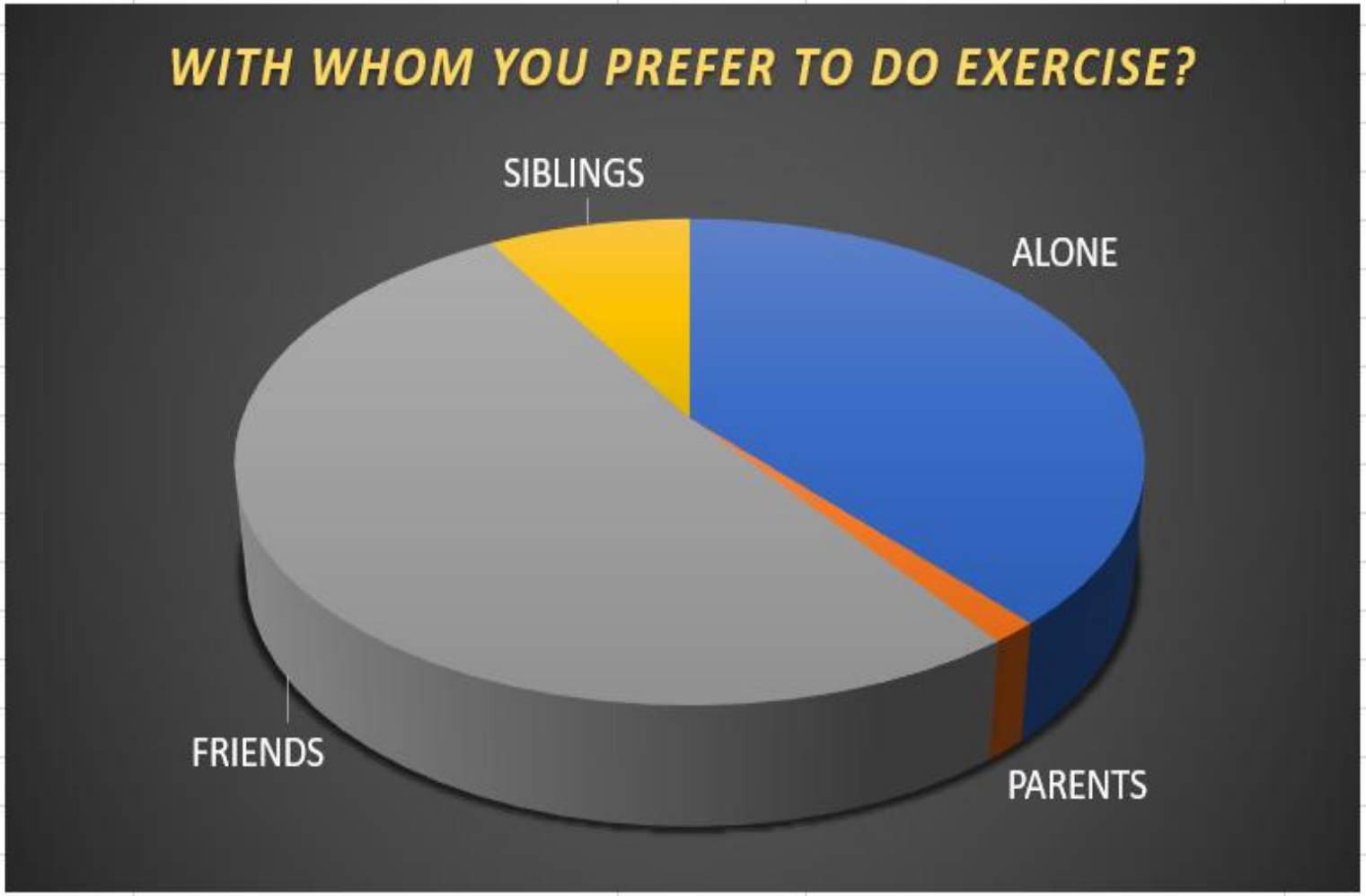
FREQUENCY FUNCTION



HOW IMPORTANT IS EXERCISETO YOU?
4
4
2
4
5
5
4
3
4
3
3
3
2
5
3
4
1
5
5
5
4
5
1
4

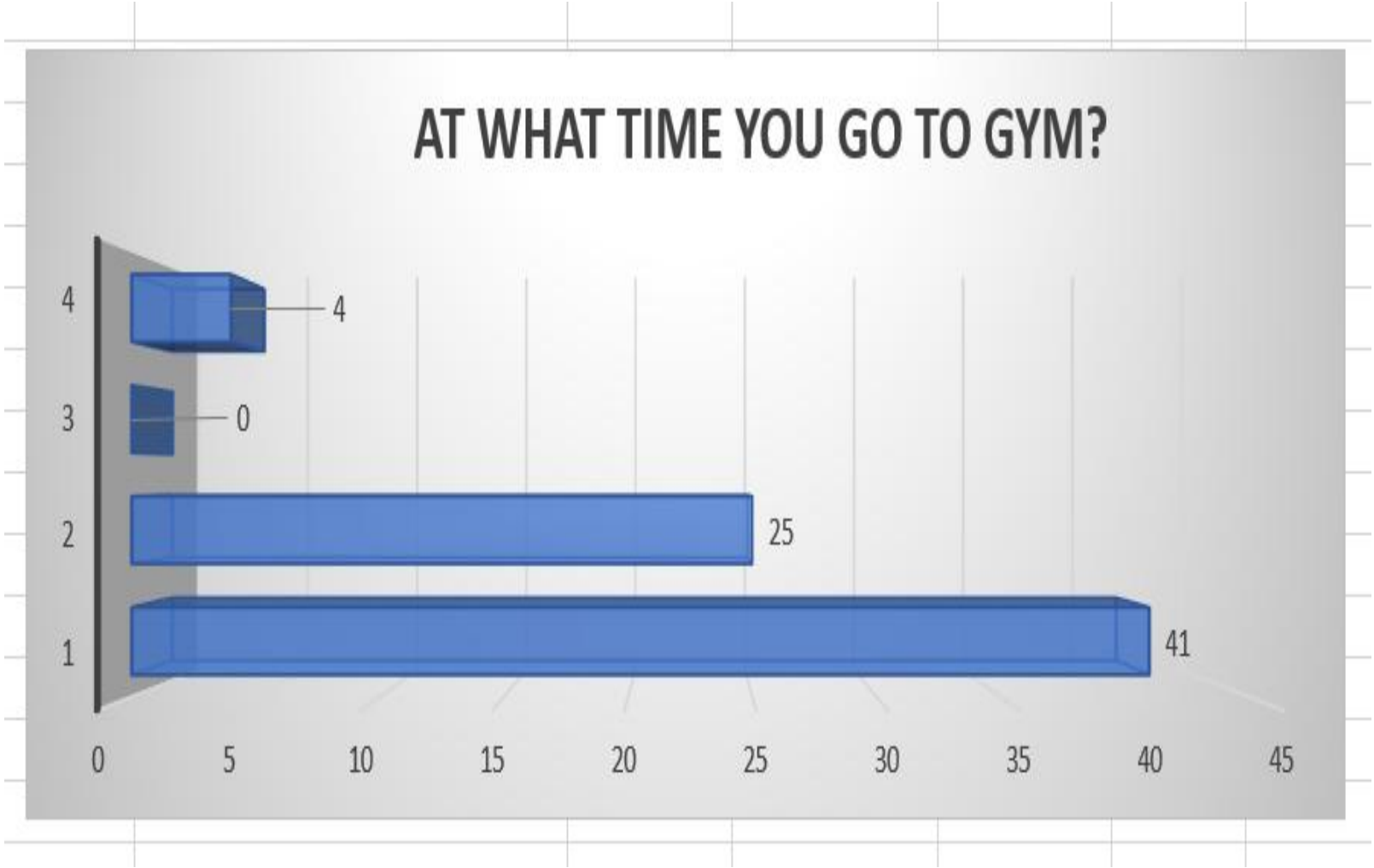
FREQUENCY	
1	7
2	7
3	14
4	23
5	19

COUNTIF FUNCTION



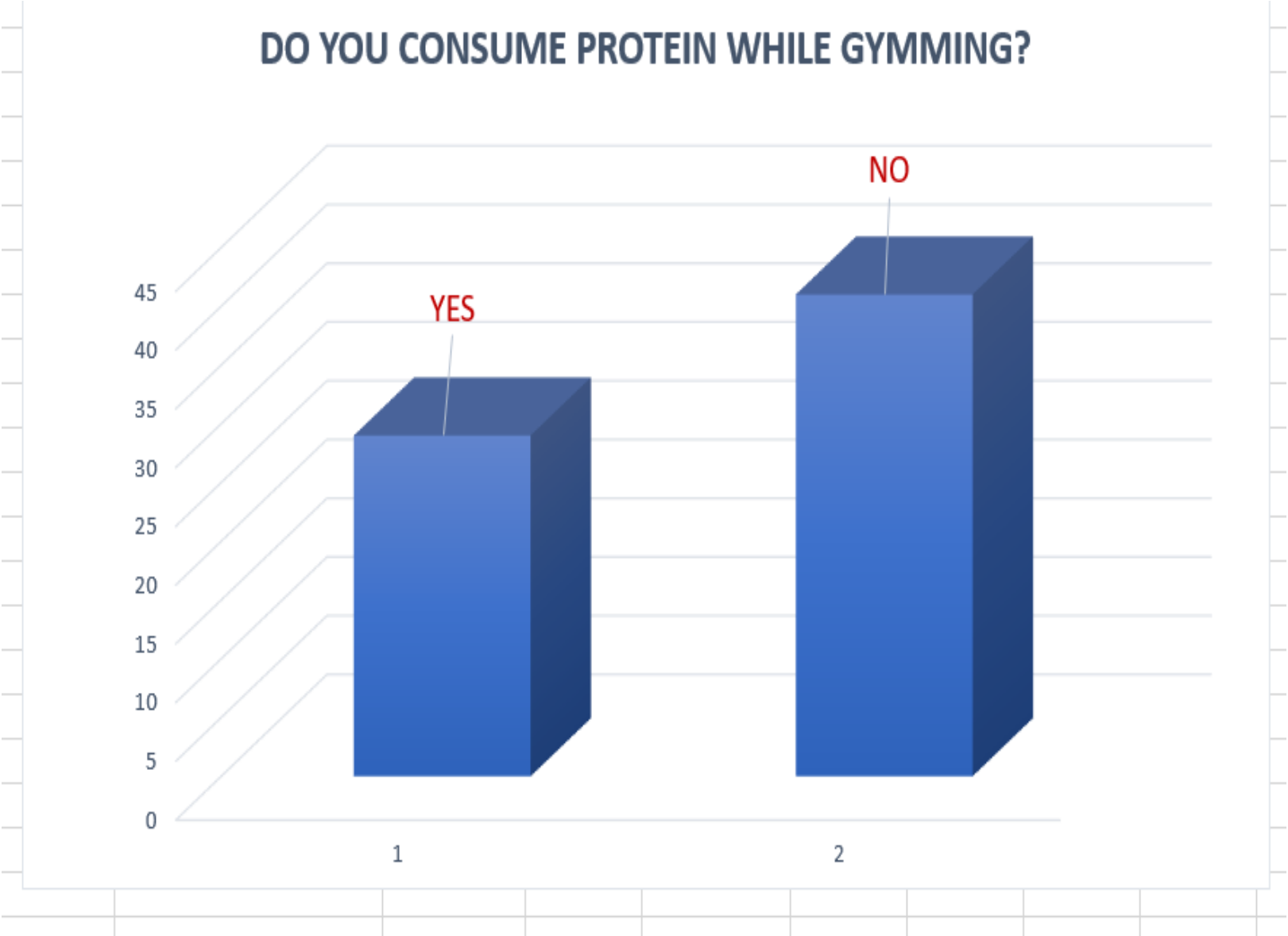
WITH WHOM YOU PREFER TO DO EXERCISE?
ALONE
ALONE
ALONE
ALONE
FRIENDS
ALONE
FRIENDS
FRIENDS
FRIENDS
ALONE
ALONE
ALONE
SIBLINGS
ALONE
FRIENDS
FRIENDS
FRIENDS
ALONE
FRIENDS
FRIENDS
ALONE
FRIENDS
FRIENDS
FRIENDS

	FREQUENCY
ALONE	27
PARENTS	1
FRIENDS	36
SIBLINGS	6
TOTAL	70



AT WHAT TIME YOU GO TO GYM?
MORNING
EVENING
EVENING
MORNING
MORNING
MORNING
MORNING
EVENING
MORNING
MORNING
MORNING
EVENING
MORNING
MORNING
MORNING
MORNING
EVENING
MORNING
NIGHT
EVENING
MORNING
MORNING
EVENING

	FREQUENCY
MORNING	41
EVENING	25
AFTERNOON	0
NIGHT	4
TOTAL	70



DO YOU CONSUME PROTEIN WHILE GYMMING?
NO
NO
NO
NO
NO
YES
NO
YES
NO
NO
NO
YES
NO
NO
NO
NO
NO
YES
YES
YES
NO
NO
NO
NO

	FREQUENCY
YES	29
NO	41

Sheet5 | Sheet1 | DESCRIPTIVE | CORRELATION | T TEST